DRUGS · REFLUX

H2 receptor antagonists

H2 receptor antagonists reduce gastric acid by blocking histamine at the parietal cell. They are less potent than PPIs but act within an hour and are useful at bedtime for nocturnal acid breakthrough, as an alternative when PPIs are not tolerated, and in some regimens for laryngopharyngeal reflux. Ranitidine was withdrawn worldwide in 2020 because of contamination with the carcinogen NDMA; famotidine is the usual choice.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

H2 RECEPTOR ANTAGONISTS

	SUMMARY
Agents	Famotidine; cimetidine; nizatidine (ranitidine withdrawn)
How they work	Competitive antagonism of histamine H2 receptors on parietal cells, reducing basal and nocturnal acid secretion
ENT uses	Nocturnal reflux symptoms with a PPI; PPI intolerance; premedication to reduce aspiration risk in some anaesthetic protocols
Adult dose	Famotidine 20 to 40 mg once or twice daily (reduce in renal impairment)
Child dose	Famotidine on specialist advice by weight

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Place in treatment

Tolerance (tachyphylaxis) develops within weeks of regular use, limiting their long-term effect. At bedtime, a dose with a morning PPI is sometimes used for nocturnal symptoms.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Headache, dizziness, diarrhoea; confusion in the elderly and in renal impairment; gynaecomastia and many interactions with **cimetidine** (a CYP inhibitor raising warfarin, phenytoin and theophylline levels).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism compared with PPIs.
- Why ranitidine was withdrawn.

4.2 Common pitfalls

COMMON PITFALLS

- Choosing cimetidine in a patient on warfarin.

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